

# A1 scalar analytic branch – the precise positivity hypothesis of A2/A3

TECT verification-first repository · claim A1-SCALAR-ANALYTIC-BRANCH

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## 1. Purpose and scope

This card isolates the exact kernel hypothesis that the analytic theorems A2 (well-posedness) and A3 (renormalisation) rely on. It is stated in terms of the shell mass  $m_{\text{sh}}^2$  (A1-KERNEL-IDENTITY), not the deprecated  $\mu^2$ . Verification: [codes/foundations/a1\\_kernel\\_checks.py](#) (a1\_kernel\_checks.py, all PASS).

## 2. The branch

**Theorem (A1-SCALAR-ANALYTIC-BRANCH).** *Assume  $Y > 0$ ,  $Z < 0$  and  $m_{\text{sh}}^2 > 0$ . Then by A1-KERNEL-IDENTITY*

$$K(q) = m_{\text{sh}}^2 + Y(|q|^2 - q_\star^2)^2 \geq m_{\text{sh}}^2 > 0 \quad \text{for all } q, \quad (1)$$

and on a fixed periodic cell  $\mathbb{T}_L^3$ ,

$$\lambda_0(L) := \min_{k \in \Lambda^*} K(k) \geq m_{\text{sh}}^2 > 0, \quad (2)$$

with equality  $\lambda_0(L) = m_{\text{sh}}^2$  only if a lattice mode lies exactly on the shell  $|k| = q_\star$  (non-generic).

**Role.** This  $m_{\text{sh}}^2 > 0$  is precisely the input used as “ $\mu^2 > 0$ ” in A2 (sectoriality:  $-L$  generates an analytic semigroup; the  $H^2$  norm equivalence) and in A3 (IR finiteness:  $G = 1/K \leq 1/m_{\text{sh}}^2$  bounded; the  $q^{-4}$  UV decay from  $Y > 0$ ). At the production anchor  $m_{\text{sh}}^2 = 5 \times 10^{-3}$ ,  $Y = 1$ ,  $Z = -0.925$ , with  $r = K(0) = 0.219$  distinct from  $m_{\text{sh}}^2$  (assert analytic\_branch\_K\_ge\_msh2\_positive, lambda0\_ge\_msh2\_equality\_onshell\_only).

## 3. Devil’s-advocate

( $\alpha$ ) “ $\lambda_0 = m_{\text{sh}}^2$  on a generic cell.” DISMISSED — only the lower bound  $\lambda_0 \geq m_{\text{sh}}^2 > 0$  is used; equality is non-generic (the verification shows  $\lambda_0 = 8.6 \times 10^{-3} > m_{\text{sh}}^2$  on a generic cell). ( $\beta$ ) “Is this the same as the old  $\mu^2 > 0$ ?” VALID and the reason for the card: the old  $\mu^2$  was the shell value  $K(q_0)$ , so  $\mu^2 = m_{\text{sh}}^2$  in the analytic notes — but the SOLVER’s  $\text{mu2} = r$  (zero momentum) is different; A2/A3 must hypothesise  $m_{\text{sh}}^2 > 0$ , not the solver’s  $\text{mu2}$ . ( $\gamma$ ) “ $Z < 0$  needed?” Yes, for the shell  $q_\star^2 > 0$  to exist. Quantitative sanity check:  $\min_q K = m_{\text{sh}}^2 = 5 \times 10^{-3}$ ;  $\lambda_0(L) = 8.6 \times 10^{-3} > m_{\text{sh}}^2$ .

## 4. Result footer

Result ID: A1-SCALAR-ANALYTIC-BRANCH  
Precise statement:  $Y > 0, Z < 0, m_{sh2} > 0 \Rightarrow K(q) = m_{sh2} + Y(q^2 - q_*^2)^2 \geq m_{sh2} > 0$  for all  $q$ ;  $\lambda_0(L) = \min_k K(k) \geq m_{sh2} > 0$ , equality only if a mode is exactly on shell.  
Scope: scalar Brazovskii kernel; the precise A2/A3 hypothesis.  
Dependencies: A1-KERNEL-IDENTITY.  
Hypotheses:  $Y > 0; Z < 0; m_{sh2} > 0$ .  
Evidence grade: PROVED CONDITIONAL (analytic) + EXECUTED (a1\_kernel\_checks.py; this card: analytic\_branch\_D\_ge\_mu2shell\_positive, lambda0\_ge\_mu2shell\_equality\_onshell\_only).  
Reproduction command: python codes/foundations/a1\_kernel\_checks.py  
Expected output: all PASS (this card: analytic\_branch\_D\_ge\_mu2shell\_positive, lambda0\_ge\_mu2shell\_equality\_onshell\_only); exit 0.  
Tier: T6 PROVED CONDITIONAL (submitted for sign-off).  
No-overclaim statement:  $m_{sh2} > 0$  (shell), NOT the solver's  $\mu_2 = r$  (zero momentum); no production-config certification (that is A1-PRODUCTION-KERNEL-MANIFEST).  
Next required action: Operator sign-off; A2/A3 depend on THIS card.